Upstox Bot: Complete Guide

What it is, how it works, the strategies, the research, and how it is deployed

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Version 1.0 · November 2026 · built by $um!t_K@Ktw@n · for Indian index and stock options (NSE / BSE) with free Upstox data
Project: github.com/Sumit211995/upstox_bot (private)

• Current mode: PAPER TRADING ONLY. No order code at all.

> **Current mode: PAPER TRADING ONLY.** This bot has **no order code at all**: it cannot place a trade on Upstox or anywhere else. It simulates every trade with real prices and the exact rules it would use with real money, and records the results. The optional Upstox token it uses is a read-only *Analytics* token that cannot place orders.

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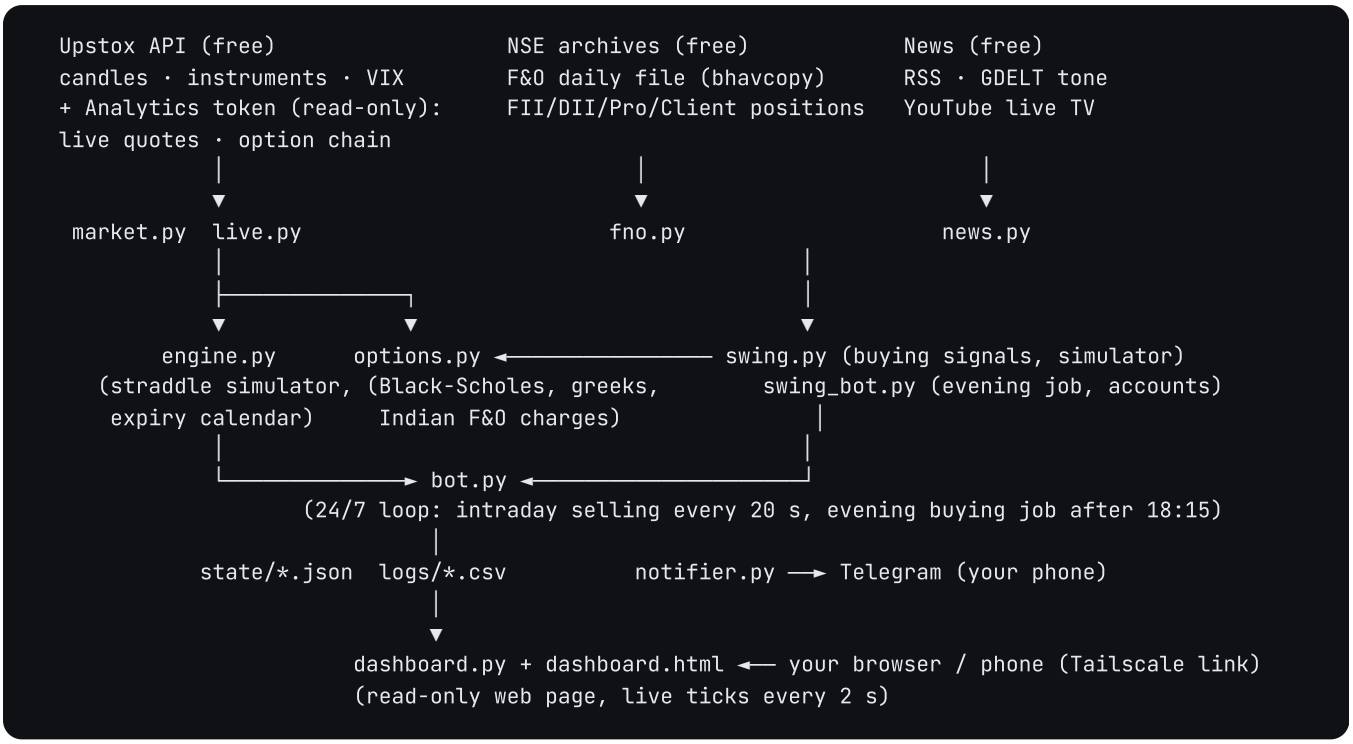
1. What the bot does, in one page

The bot paper-trades Indian options in **two separate ways**, because they need very different amounts of money.

	Selling (intraday)	**Buying** (swing, 1–8 days)
What	Sell the at-the-money call + put (a **straddle**) and buy cheap **wings** 10 strikes away as insurance	Buy a call (CE) or put (PE), or a cheap **debit spread**, when one of four tested signals fires
Where	Nifty, Sensex, Bank Nifty, Midcap Nifty	Nifty and Bank Nifty (+ 20 large stocks with monthly options)
When	Enter 10:00, stop per leg, exit 15:15 the same day	Signal after the close (evening), buy at the next morning's open, exit on stop / target / time
Money needed	Large margin (₹1 lakh+ per lot even hedged)	Starts at **₹20,000**
Why it earns	Option premium decays to near zero by expiry; wings cap the bad days	Follows where the big money (FII/DII) is positioned, plus breakouts / trend starts
Records	6 paper "books" (`books.json`)	2 paper accounts, ₹20k and ₹50k (`buy_books.json`)

The bot does **not** predict the future. Each rule was chosen on 2022–2024 data and then checked on 2025–2026 data it never saw. Only rules that made money in **both** periods were kept. Everything is still **paper** until real results match the research (see chapter 16).

2. The big picture: parts and how they connect



Two programs, one set of files. `bot.py` simulates the trades and writes `state/` and `logs/`. `dashboard.py` only **reads** those files plus public prices, so the web page can never change anything. On the server each runs in its own Docker container and both share the same folders.

2.1 Tech stack at a glance

Layer	Technology	Where
Language	**Python 3.11** (bot, research, tests, web server) · HTML/CSS/JavaScript (dashboard page only)	all `*.py`, `dashboard.html`
Data and maths	**pandas**, **NumPy**	everywhere
Option pricing	**Black-Scholes** model, implied volatility, greeks (Δ, Γ, θ, vega); real Indian F&O charges	`options.py`
Market data	**Upstox public API v3** (candles, instrument list); **Upstox Analytics token** for live quotes + option chain	`market.py`, `live.py`
Exchange files	**NSE bhavcopy** (daily option prices, OI), **participant-wise OI** (FII/DII/Pro/Client)	`fno.py`
News	RSS (Economic Times, Moneycontrol, Business Standard, Mint, CNBC-TV18, NDTV Profit, BusinessLine), **GDELT** tone, YouTube live	`news.py`
Storage	Plain **JSON** (state) and **CSV** (journals) + **pickle** caches; no database	`state/`, `logs/`, `data/`
Web dashboard	Python's built-in `http.server` + **TradingView Lightweight Charts**; read-only, strict security headers	`dashboard.py`, `dashboard.html`

Layer	Technology	Where
Alerts	**Telegram Bot API** (same bot as delta_bot)	`notifier.py`
Tests	**pytest**, 28 tests	`tests/`
Hosting	**Docker + Docker Compose** on **Oracle Cloud Always Free** (Mumbai), private access through **Tailscale**	`Dockerfile` , `docker-compose.yml`
Source	**Git + GitHub** (private repo `upstox_bot`)	

File	Lines	Job
`bot.py`	~255	Main loop. Every 20 s in market hours replays each selling book on today's candles; after 18:15 runs the buying evening job.
`engine.py`	~415	Selling simulator (hedged straddle, per-leg stop), expiry calendar, older intraday buying families (research).
`options.py`	~145	Black-Scholes, implied volatility, greeks, model volatility from VIX, Indian F&O charges.
`market.py`	~205	Upstox candles (5-minute history since 2022, cached), intraday, instrument list, India VIX daily.
`live.py`	~160	Live quotes and option chain with the Analytics token; live estimate for open buy positions without it.
`fno.py`	~245	NSE F&O daily files and participant positions (FII/DII/Pro/Client), cached.
`swing.py`	~650	Buying features (OI, smart money, EMA, RSI, ADX, breakouts, VIX), the signal families, the trade simulator, accounts.
`swing_bot.py`	~350	The evening job: builds tonight's signals and contracts, updates the paper accounts, journals, Telegram.
`news.py`	~210	Headlines per channel / symbol, mood score, GDELT tone, live TV video ids.
`dashboard.py` + `dashboard.html`	~440 + ~1,210	The web page and its read-only API (`/api/live` , `/api/buying` , `/api/chart` , ...).
`config.py` , `notifier.py`		Settings from `.env` ; Telegram.
`books.json` , `buy_books.json`		The selling books and the buying strategies / accounts.
`research/`		Every research script and its results (`research/out/`).

3. One day of the bot, hour by hour

Time (IST)	What happens
09:15	Market opens. Dashboard prices update every **2 seconds**. Yesterday's buy signals are bought at the **opening price**.
09:15–15:15	Open buy positions are marked live (real option price with the token, an estimate without it).
10:00	Selling books enter: sell ATM CE + PE, buy wings 10 strikes away (only on the days each book trades).
10:00–15:15	Every 20 s: check each short leg against its stop. A stopped leg is closed; the other leg stays.
15:15	Selling books close everything. Indices freeze for the **closing auction** (since Aug 2026) until ~15:30.
15:30	Market closes.
18:15 →	Evening job (retries every 15 minutes until NSE has published the day's files, last try 23:45): downloads the F&O file and FII/DII positions, updates the buying accounts (stops, targets, time exits on real prices), computes **tonight's signals** and the exact contracts to buy tomorrow, sends them to Telegram.

4. Selling strategy: hedged short straddle

The trade. At 10:00 sell the **at-the-money** call and put of the nearest expiry and buy the call and put **10 strikes further out** (the wings). You collect premium; the wings cap the loss if the index runs far in one direction.

The exits. Each short leg has its own stop: if its premium rises by the book's stop % (60–100%), that leg is bought back. At 15:15 everything is closed.

Why it works. An option's time value melts fastest on expiry day. On most days the index moves less than the premium collected. Stops + wings keep the rare big trend day from wiping out many good days.

Book	Days traded	Stop per leg
Nifty · expiry day	Weekly expiry (Tuesday)	80%
Nifty · every day	Every day of the weekly cycle	100%
Sensex · expiry day	Weekly expiry (Thursday, BSE)	60%
Sensex · every day	Every day of the weekly cycle	80%
Bank Nifty · expiry day	Monthly expiry only (≈ 1 trade a month)	100%
Midcap Nifty · expiry day	Monthly expiry only	100%

5. Buying strategy: four signals for a small account

All buying signals are computed **after the close** from the day's final data, and bought at the **next morning's open**. This removes any chance of using information that was not yet available.

5.1 The signals (in priority order)

#	Signal	Rule	Contract	Stop · target · max days
1	**Index smart money**	Average 60-day rank of five NSE participant measures (DII futures share, DII futures change, FII options net long, *against* DII options net, *against* retail futures change). Top 20% → CE, bottom 20% → PE.	Nifty / Bank Nifty, ATM, ≥ 25 days to expiry	−50% · none · 5
2	**Index 20-day breakout (1:2)**	Close above the highest high of 20 days → CE; below the lowest low → PE.	1 strike in the money	−35% · +70% · 8
3	**Index EMA 9/21 cross (1:2)**	EMA 9 crosses EMA 21 in the direction of EMA 50.	1 strike in the money	−25% · +50% · 3
4	**Stock smart money**	The same market-wide score applied to 20 large F&O stocks. **Skipped when India VIX is in the top 20% of its last 60 days** (options too expensive).	Monthly stock option, 1 strike ITM	−30% · none · 5

Shown on the dashboard but **not traded**: the **OI-extreme** signal (the original "30:70 open-interest" idea, measured as a 60-day extreme of call:put OI within ±2% of the price). It works only with one exact setting, so it is not trusted.

5.2 Why "smart money" works

NSE publishes every evening how many futures and options contracts each group holds: **FII** (foreign funds), **DII** (Indian funds), **Pro** (brokers' own trading) and **Client** (retail). Research found that following DII futures and FII options, and going *against* retail futures, pointed the right way over the next days. One group alone did not work (FII futures alone: PF 0.70); the five-part score did (PF 3.85 tuning, 4.06 unseen).

5.3 Greeks: do we need them?

The cards show **θ (theta)**, what the position loses per day just from time passing, and **Δ (delta)**, how much it moves per 1 point of the index. They are information, not rules. The ₹20k account uses **debit spreads** mainly because they cut theta: e.g. a Nifty PE lost −₹790/day as a plain buy vs −₹138/day as a 6-strike spread.

6. Money: the ₹20k and ₹50k paper accounts, with a worked example

Both accounts take signals in priority order, at most **2 positions at once**, and only if the cash is there.

Account	Capital	Per trade	Index signals bought as
₹20k (recommended start)	₹20,000	≈ ₹7,000	**Debit spread**: buy ATM/ITM, sell the option 6 strikes further out
₹50k	₹50,000	≈ ₹15–20k	Plain CE / PE

Worked example (₹20k, illustrative numbers). Smart-money score hits its top 20% on Nifty in the evening → tomorrow buy Nifty 25,300 CE and sell 25,600 CE (6 strikes × 50). Say the spread costs ₹110 × 65 units = ₹7,150. The stop is −50% (spread worth ₹55 → loss ≈ ₹3,600 + charges). The trade is closed after 5 trading days at the latest. The maximum the spread can ever be worth is the width, 300 points × 65 = ₹19,500.

7. Research results: what the data says

Method. Every trade was simulated as a real option trade: real NSE daily option prices (OHLC) for buying, modelled premiums checked against NSE prices for intraday selling, lot sizes, brokerage, STT, exchange + SEBI charges, stamp duty, GST and slippage (0.5% index, 1.5% stocks). **Settings chosen on 2022–2024, judged on 2025–2026.** If both stop and target were touched on the same day, the **stop is assumed first** (pessimistic).

7.1 Selling books (1 lot)

Book	2022–24: trades · win · PF	2025–26 (unseen): trades · win · PF · net
Nifty · expiry day	151 · 60% · 2.02	**101 · 58% · 2.50 · +₹1.25 L**
Nifty · every day	723 · 69% · 1.91	**461 · 66% · 1.96 · +₹2.56 L**
Sensex · expiry day	80 · 70% · 2.37	**91 · 76% · 1.66 · +₹65k**
Sensex · every day	405 · 60% · 1.79	**461 · 56% · 1.44 · +₹95k**
Bank Nifty · expiry day	155 · 52% · 2.47	**29 · 59% · 3.48 · +₹62k**
Midcap Nifty · expiry day	149 · 68% · 4.75	**24 · 71% · 2.23 · +₹39k**

7.2 Buying signals (each alone)

Signal	2022–24: trades · win · PF	2025–26: trades · win · PF
Index smart money	18 · 67% · 3.85	**11 · 55% · 4.06**
Index 20-day breakout (1:2)	36 · 53% · 2.27	**20 · 50% · 2.03**
Index EMA cross (1:2)	15 · 60% · 3.01	**15 · 40% · 1.31**
Stock smart money (VIX filter)	192 · 51% · 2.36	**117 · 50% · 1.95**
OI-extreme (shown only)	149 · 37% · 1.92	73 · 32% · 1.67

7.3 Combined accounts

Account	2025–26 (unseen)	Start in any month, run 12 months (47 runs)
₹20k (spreads)	37 trades · 60% win · PF 2.1 · ₹20k → ₹58k · lowest ₹15.3k	profit in **74%** · median end ₹27.4k · fell below half 9%
₹50k	46 trades · 48% win · PF 2.4 · ₹50k → ₹2.38 L	profit in **87%** · median end ₹1.18 L · fell below half 6%

> **Read these numbers honestly.** Index smart money has only **11** unseen trades: its PF could be much lower in reality. One in four 12-month runs of the ₹20k account still lost money. Paper first.

8. What did not work (tested and rejected)

Idea	Result
Intraday option **buying** (opening-range breakout, Supertrend, pullback) on any index	Lost in both periods on all 4 indices (unseen PF 0.5–0.8)
Plain stock option buying on price signals	Lost in ~99.6% of settings
A fixed **30:70 OI ratio**	Almost never happens on indices; the relative (60-day) version is only shown
FII futures long % alone	PF 0.70
News as a filter	Made results worse (PF 1.44 → 1.13); news tone is shown as context only
RSI, MACD, Supertrend, Bollinger squeeze as buying signals	Failed on unseen data
VIX filter on index buys	No setting improved both periods
Day filters for selling (ADX, RSI, smart money, squeeze, VIX)	None improved both periods without cutting total profit
Full-lot buying with ₹20k	Only 47% of 12-month runs profitable → debit spreads instead

9. Data sources (all free)

Data	Source	Used for
Index candles (5-minute, since Jan 2022)	Upstox public historical API	Selling simulator, charts
Live quotes, option chain + greeks	Upstox **Analytics token** (read-only, 1 year, free)	Dashboard ticks, live marks
Daily option prices + open interest	NSE F&O bhavcopy	Buying research and the evening job
FII / DII / Pro / Client positions	NSE participant-wise OI	Smart-money score
FII / DII cash buying	NSE (latest day)	Big money panel
India VIX	Upstox	Stock-buy filter, option model
Headlines	RSS feeds of 7 business outlets	News panel
News tone	GDELT	Context only
Live business TV	YouTube (CNBC-TV18, ET Now, NDTV Profit, Zee Business, CNBC Awaaz)	News & TV tab

How to get the Upstox Analytics token: log in at the Upstox developer portal → *Apps* → **Analytics token** → generate (valid one year) → paste it as `UPSTOX_ANALYTICS_TOKEN=` in `.env` on the server and restart. It is free and read-only.

10. The dashboard

Tabs: **Overview · Buying · Selling · Big money · News & TV.**

- **Price tiles** (Nifty, Bank Nifty, Sensex, Midcap, VIX) update every 2 seconds and flash on each change. The pill shows *Live · Upstox* (real quotes) or *Live · ~1 min* (no token). From 15:15 to 15:30 it shows the closing-auction pause.
- **Chart** (5m / 15m / 1h / 1D): the last candle moves with the live price. The Buying tab shows buy arrows from the top 3 buying strategies (Nifty, daily); the Selling tab shows the straddle trades of the top selling books.
- **Signal cards**: every buy and sell signal with contract, stop, target, θ, Δ and the strategy's success rate, best first.
- **Buying tab**: the ₹20k / ₹50k accounts, tonight's signals, ranked strategies, all trades (paper and backtest), OI scanner.
- **Big money**: FII / DII / Pro / Client positions and their 60-day ranks; FII / DII cash.
- **News & TV**: headlines per channel and live TV inside the page.

11. Telegram alerts

The same Telegram bot as delta_bot. Messages: bot started, selling entries / stops / exits, the evening buying plan (contract, stop, target), buy fills and exits.

12. Security

Rule	How
No trading possible	There is no order code; the token is read-only
Secrets stay private	`.env` is never committed (`.gitignore`), `chmod 600` on the server, the token is only sent to `api.upstox.com`
Page cannot be attacked from outside	Bound to `127.0.0.1` ; reachable only through your private Tailscale network
Page hardening	Content-Security-Policy, `X-Frame-Options: DENY` , `nosniff` , only GET allowed, parameter allowlist, http(s)-only links

13. Deployment: Oracle Cloud + Tailscale

The bot runs next to delta_bot on the same **Oracle Cloud Always Free** VM (Mumbai, 1 GB RAM + 2 GB swap).

	delta_bot	upstox_bot
Folder	`~/delta_bot`	`~/upstox_bot`
Containers	`delta-bot`, `dashboard`	`upstox-bot`, `upstox-dashboard`
Dashboard (on the server)	`127.0.0.1:8000`	`127.0.0.1:8002`
Link (phone / laptop with Tailscale)	`https://<your-server>.<tailnet>.ts.net`	**`https://<your-server>.<tailnet>.ts.net:8443`**

Open it on your phone: install the Tailscale app, sign in with the same account, open the link.

Update after a code change:

```
# on the laptop
git push
# on the server
ssh -i ~/Downloads/oracle.key ubuntu@<server-ip>
cd ~/upstox_bot && git pull && docker compose up -d --build
```

Memory. The two Upstox containers use ≈ 230 MB. The evening job briefly needs more and runs ~10–15 minutes on the small server (it uses swap). Never start a second evening job by hand while the bot's own one is running.

Where to check it is running: the Oracle console only shows the VM (CPU / memory graphs). The bots themselves: the dashboard link, Telegram "bot started" messages, or on the server `docker compose ps` and `docker compose logs --tail 30`.

14. Daily use: commands

Command	What it does
`python bot.py`	Run the bot (loop)
`python bot.py --status`	Paper results so far
`python bot.py --evening`	Run the buying evening job now (only if the bot is not already running it)
`python dashboard.py`	Dashboard on http://localhost:8002
`python -m pytest -q`	Run the 28 tests
`python research/buy_final.py`	Rebuild the buying history shown on the dashboard

15. Configuration reference (`.env`)

Key	Default	Meaning
`INDICES`	`NIFTY,BANKNIFTY,SENSEX,MIDCPNIFTY`	Indices shown and traded by the selling books
`PAPER_CAPITAL`	`200000`	Paper capital per selling book (₹)
`LOTS`	`1`	Lots per selling trade
`POLL_SECONDS`	`20`	Check interval in market hours
`TELEGRAM_TOKEN` , `TELEGRAM_CHAT_ID`	empty	Telegram alerts (optional)
`UPSTOX_ANALYTICS_TOKEN`	empty	Live quotes + option chain (optional, read-only)

Put comments on **their own line** in `.env` , never after an empty value (`KEY= # note`): Docker would read the comment as the value.

16. The plan: from paper to real money

1. **Paper for 1–2 months** (now). Watch the ₹20k account and the selling books on the dashboard.
2. **Compare with the research.** Go further only if paper results look roughly like chapter 7: at least ~15 buy trades, profit factor above ~1.3, no rule surprises.
3. **Start small.** Real buying with ₹20k using only the signals that held up on paper; selling only with enough margin.
4. **Real orders are not built.** Placing orders would need Upstox's order API with a separate trading token, plus the same safety layer as delta_bot (daily loss limit, kill switch). That is a deliberate later step.

17. Troubleshooting and FAQ

Question	Answer
Index price stops moving at 15:15	Normal since Aug 2026: closing auction. VIX keeps moving.
No buy signals tonight	Normal: the signals fire a few times a month.
Why no buy signals on Sensex / Midcap?	Buying research used NSE data (Nifty, Bank Nifty); Sensex is on BSE.
Dashboard says "Live · ~1 min"	The Upstox token is missing or expired (one year); generate a new one.
The evening job did not run	NSE publishes late some days; the bot retries every 15 minutes until 23:45.
Server CPU near 100% in the evening	The evening job; it finishes in ~15 minutes.

18. Glossary

Term	Meaning
CE / PE	Call option (profits when the index rises) / put option (profits when it falls)
ATM / ITM / OTM	Strike at / inside / outside the current price
Straddle	Selling (or buying) a CE and a PE at the same strike
Wings	Far-away options bought as insurance on a short straddle
Debit spread	Buy one option, sell a cheaper one further out: lower cost and decay, capped profit
Expiry	Last day of an option contract
OI	Open interest: contracts still open
FII / DII / Pro / Client	Foreign funds / Indian funds / brokers' own trading / retail
PF (profit factor)	Total won ÷ total lost. Above 1 = profitable
Walk-forward	Choose settings on old data, judge on newer unseen data
VIX	India's volatility index: high = options expensive
θ theta / Δ delta	Daily time decay / move per 1 point of the underlying
Bhavcopy	NSE's daily file of every contract's prices and open interest